

Code Layout, Readability and Reusability

Date: 30 September 2026
Team ID: SWTID-2026-3656
Project Name: EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks: 5 Marks

Code Quality Checklist

No.	Parameter	Followed	Remarks
1	Consistent indentation	Yes	PEP 8-style Python
2	Proper file structure	Yes	Five modules, templates, static, configuration
3	Meaningful variables	Yes	Names describe concepts, questions, topics, and results
4	Function names	Yes	Functions express their purpose
5	Comments	Yes	Configuration and model choice are documented
6	Modular design	Yes	One module per learning feature
7	No redundant code	Yes	Shared Gemini generation helper
8	Error handling	Yes	Validation and safe endpoint responses

Reusable Components

Component	Technology	Reuse	Level
_clean_input	Python	All form endpoints	High
_error_response	Python	All route failures	High
generate_gemini_text	Python/google-genai	Q&A, quiz, summary, path	High
_load_model	Python/lru_cache	Concept explanation requests	High
formatResult	JavaScript	Text and quiz rendering	Medium

Quality Assessment

Aspect	Rating	Comment
Code layout	5/5	Matches required modular architecture
Readability	5/5	Beginner-friendly names and flow
Reusability	5/5	Shared helpers avoid duplication
Documentation	4/5	Project guide and comments included
Overall	5/5	Maintainable MVP